SECTION 4: SYMMETRIC VS ASYMMETRIC ENCRYPTION [10 MARKS]

8. Symmetric Encryption Vulnerability: (3 Marks)

Explain why symmetric encryption suffers from the "Key Distribution Problem" and describe how hackers exploit it:

Model Answer: In symmetric encryption, the same key encrypts and decrypts data. To communicate, the sender must send the secret key to the receiver across the network. If an attacker intercepts the transmission of the key, they can decrypt all future confidential messages effortlessly. [3 marks]

9. Asymmetric Public/Private Key Encryption Scenario: (4 Marks)

Zayan wishes to send a confidential, encrypted contract to Nadia over the Internet using asymmetric encryption.

a) Which cryptographic key does Zayan use to encrypt the contract? **Nadia's Public Key [1 mark]**

b) Which key does Nadia use to decrypt and read the contract? **Nadia's Private Key [1 mark]**

c) Can an eavesdropper who intercepts the message decrypt it using Nadia's Public Key? Explain why: (2 Marks)

Answer: No. In asymmetric cryptography, data encrypted with a public key can ONLY be decrypted by the matching private key, which is kept strictly secret on Nadia's device. [2 marks]

10. Digital Signatures & SSL/TLS: (3 Marks)

a) What two security properties does a Digital Signature provide? (2 Marks)

1. Authenticity: Confirms the identity of the true sender.

2. Integrity / Non-repudiation: Proves message was not altered in transit. [2 marks]

b) State the role of a Certificate Authority (CA) in issuing SSL certificates for websites: (1 Mark)

Answer: A trusted third-party organization that verifies the website owner's identity and digitally signs their certificate to prevent spoofing. [1 mark]